Unit I: Descriptive Statistics

Measures of Central Tendency, Dispersion,
Skewness & Kurtosis

STA201 – Theory of Probability and Statistics

B.Tech. Semester III (Batch 2025–2029)

July – December 2026

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1 Introduction to Descriptive Statistics

Descriptive statistics provide methods to summarize, organize, and present data in a meaningful way. For an engineer, they are the first step in understanding measurement variability, product quality, and process behaviour. The three key aspects we quantify are:

- **Central tendency** – a typical or representative value.
- **Dispersion** – the spread or variability.
- **Shape** – symmetry (skewness) and tail heaviness (kurtosis).

1.1 Types of Data

Data can be classified as:

- **Qualitative (categorical)**: non-numeric labels (e.g., machine status: on/off, type of defect).
- **Quantitative (numerical)**: numeric measurements.
  - *Discrete*: countable (e.g., number of defective items, number of cylinders).
  - *Continuous*: measurable on a continuum (e.g., voltage, temperature, diameter).

1.2 Listing of Data

Data can be presented in two fundamental ways:

1. **Raw (ungrouped) data**: a simple list of individual observations.
2. **Frequency distribution**: a table that shows distinct values (or class intervals) and how often they occur. This summarises large datasets.

In this unit, we work with both forms, using appropriate formulas for each.

2 Measures of Central Tendency

A measure of central tendency is a single value that attempts to describe the centre of a data set.

2.1 Arithmetic Mean

2.1.1 For Individual Observations

For n values $x_1, x_2, \dots, x_n$,

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i. \tag{1}$$

2.1.2 For Discrete Frequency Distributions

If values x_i occur with frequencies f_i ($i = 1, \dots, k$) and $N = \sum f_i$,

$$\bar{x} = \frac{\sum_{i=1}^{k} f_i x_i}{N}. \tag{2}$$

2.1.3 For Continuous Grouped Data

When data are grouped into class intervals, let x_i be the midpoint of the i-th class and f_i its frequency. The mean is calculated using the same formula as above.

Shortcut (Assumed Mean) Method Choose an assumed mean A (often a central midpoint). Define $d_i = x_i - A$. Then

$$\bar{x} = A + \frac{\sum f_i d_i}{N}. \tag{3}$$

Step-Deviation Method If all class widths are equal to h, set $u_i = \dfrac{x_i - A}{h}$. Then

$$\bar{x} = A + h \cdot \frac{\sum f_i u_i}{N}. \tag{4}$$

Weighted Arithmetic Mean When observations carry different weights w_i,

$$\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}. \tag{5}$$

Solved Example

The breaking strength (in MPa) of 12 identical steel rods tested in a lab is:

$$485,\ 492,\ 500,\ 495,\ 488,\ 505,\ 498,\ 490,\ 502,\ 489,\ 500,\ 491.$$

Calculate the mean tensile strength. If the design requires a mean strength of at least 495 MPa, does this batch meet the requirement?

Solution:

$$\begin{aligned}\bar{x} &= \frac{485 + 492 + 500 + 495 + 488 + 505 + 498 + 490 + 502 + 489 + 500 + 491}{12} \\ &= \frac{5935}{12} \approx 494.58 \text{ MPa}.\end{aligned}$$

The mean is slightly below 495 MPa, so the batch does **not** meet the requirement.

Solved Example

The diameters (in mm) of 100 ball bearings are summarised below.

Diameter (mm)	Frequency
10 – 12	6
12 – 14	18
14 – 16	34
16 – 18	28
18 – 20	14

Using the step-deviation method with assumed mean $A = 15$ (midpoint of 14–16), compute the mean diameter.

Solution: $h = 2$. Midpoints: 11, 13, 15, 17, 19.

$$u_i = \frac{x_i - 15}{2} : \quad -2, -1, 0, 1, 2.$$

x_i	f_i	u_i	$f_i u_i$
11	6	-2	-12
13	18	-1	-18
15	34	0	0
17	28	1	28
19	14	2	28
	$N = 100$		$\sum f_i u_i = 26$

$$\bar{x} = A + h\frac{\sum f_i u_i}{N} = 15 + 2 \cdot \frac{26}{100} = 15 + 0.52 = 15.52 \text{ mm}.$$

2.2 Median

The median is the middle value of ordered data.

2.2.1 Ungrouped Data

For n odd, median $= x_{(\frac{n+1}{2})}$; for n even, median $= \frac{1}{2}\big(x_{(\frac{n}{2})} + x_{(\frac{n}{2}+1)}\big)$.

2.2.2 Grouped Continuous Data

$$\text{Median} = L + \left(\frac{\frac{N}{2} - F}{f_m}\right) \times w,$$

where L = lower true boundary of median class, F = cumulative frequency before median class, f_m = frequency of median class, w = class width.

Solved Example

For the ball bearing data in Example 2.1.3, find the median diameter.

Solution: Cumulative frequencies:

Class	f	Cum.
$10 - 12$	6	6
$12 - 14$	18	24
$14 - 16$	34	58
$16 - 18$	28	86
$18 - 20$	14	100

$N/2 = 50$. The median class is 14–16 (cum. freq. 58 exceeds 50). $L = 14$, $F = 24$,

$f_m = 34$, $w = 2$.

$$\text{Median} = 14 + \left(\frac{50 - 24}{34}\right) \times 2 = 14 + \frac{26}{34} \times 2 \approx 14 + 1.529 = 15.529 \text{ mm.}$$

2.3 Mode

The mode is the most frequent value.

2.3.1 Grouped Continuous Data

$$\text{Mode} = L + \left(\frac{f_m - f_1}{2f_m - f_1 - f_2}\right) \times w,$$

where L = lower boundary of modal class, f_m = frequency of modal class, f_1 = frequency of class before, f_2 = frequency of class after.

Solved Example

Find the modal diameter for the ball bearing data.

Solution: The highest frequency is 34 (class 14–16). Thus $L = 14$, $f_m = 34$, $f_1 = 18$, $f_2 = 28$, $w = 2$.

$$\text{Mode} = 14 + \left(\frac{34 - 18}{2 \cdot 34 - 18 - 28}\right) \times 2 = 14 + \left(\frac{16}{68 - 46}\right) \times 2 = 14 + \frac{16}{11} \approx 15.455 \text{ mm.}$$

2.4 Empirical Relationship (Mean, Median, Mode)

For a moderately skewed unimodal distribution:

$$\text{Mean} - \text{Mode} \approx 3(\text{Mean} - \text{Median}).$$

3 Measures of Dispersion

Dispersion indicates the spread or variability. An engineer must know how consistent a process is.

3.1 Range

$$R = X_{\max} - X_{\min}.$$

Coefficient of Range $= \dfrac{X_{\max} - X_{\min}}{X_{\max} + X_{\min}}$.

3.2 Mean Deviation (Average Absolute Deviation)

$$MD_{\text{mean}} = \frac{\sum f_i |x_i - \bar{x}|}{N}, \qquad MD_{\text{median}} = \frac{\sum f_i |x_i - \text{Med}|}{N}.$$

3.3 Standard Deviation and Variance

Variance is the average squared deviation from the mean.

3.3.1 Population Standard Deviation

Individual observations:

$$\sigma = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2}. \tag{6}$$

Frequency distribution:

$$\sigma = \sqrt{\frac{\sum f_i(x_i - \bar{x})^2}{N}}. \tag{7}$$

3.3.2 Shortcut (Assumed Mean) Formula

Let $d_i = x_i - A$. Then

$$\sigma = \sqrt{\frac{\sum f_i d_i^2}{N} - \left(\frac{\sum f_i d_i}{N}\right)^2}. \tag{8}$$

3.3.3 Step-Deviation Formula

For equal class width h, $u_i = \frac{x_i - A}{h}$,

$$\sigma = h \cdot \sqrt{\frac{\sum f_i u_i^2}{N} - \left(\frac{\sum f_i u_i}{N}\right)^2}. \tag{9}$$

3.3.4 Coefficient of Variation (CV)

$$CV = \frac{\sigma}{\bar{x}} \times 100\%.$$

Solved Example

A manufacturer records the lifetimes (hours) of 10 LED bulbs: 1010, 980, 1005, 1020, 990, 1015, 985, 1000, 995, 1010. Compute the standard deviation and coefficient of variation. Interpret the consistency.

Solution: Mean $\bar{x} = 1001$ hrs. Deviations: $9, -21, 4, 19, -11, 14, -16, -1, -6, 9$.

$$\sum(x_i - \bar{x})^2 = 81 + 441 + 16 + 361 + 121 + 196 + 256 + 1 + 36 + 81 = 1590.$$

$\sigma = \sqrt{1590/10} \approx 12.61$ hrs. $CV = (12.61/1001) \times 100\% \approx 1.26\%$. The low CV indicates very consistent lifetimes.

Solved Example

Use the step-deviation method to compute the standard deviation of the ball bearing diameters from Example 2.1.3.

Solution: We already have $A = 15$, $h = 2$, $u_i = -2, -1, 0, 1, 2$, $\sum f_i u_i = 26$. Now compute $f_i u_i^2$:

x_i	f_i	u_i	$f_i u_i^2$
11	6	-2	24
13	18	-1	18
15	34	0	0
17	28	1	28
19	14	2	56
	100		$\sum f_i u_i^2 = 126$

$$\sigma = h\sqrt{\frac{\sum f_i u_i^2}{N} - \left(\frac{\sum f_i u_i}{N}\right)^2} = 2\sqrt{\frac{126}{100} - \left(\frac{26}{100}\right)^2} = 2\sqrt{1.1924} \approx 2.18 \text{ mm}.$$

4 Moments

Moments provide a systematic way to describe the shape of a distribution. The r-th moment about an arbitrary value A is defined as

$$m'_r = \frac{\sum f_i (x_i - A)^r}{N}.$$

When $A = \bar{x}$, we obtain the **central moments** (moments about the mean):

$$m_r = \frac{\sum f_i (x_i - \bar{x})^r}{N}. \tag{10}$$

The first central moment m_1 is always zero. Important special cases:

- $m_2 = \sigma^2$ (variance).
- m_3 is related to skewness.
- m_4 is related to kurtosis.

With central moments defined, we now use them to quantify skewness and kurtosis.

5 Skewness

Skewness measures the asymmetry of a distribution. A symmetric distribution has zero skewness. Positive skewness means the right tail is longer (mean $>$ median $>$ mode), negative skewness means the left tail is longer.

5.1 Pearson's Coefficient of Skewness

Based on the empirical differences among mean, median, and mode:

$$Sk_P = \frac{\text{Mean} - \text{Mode}}{\sigma} \quad \text{or} \quad Sk_P = \frac{3(\text{Mean} - \text{Median})}{\sigma}.$$

The sign indicates direction, magnitude indicates degree of asymmetry.

5.2 Moment Coefficient of Skewness

Using the third central moment:

$$\beta_1 = \frac{m_3^2}{m_2^3}, \qquad \gamma_1 = \sqrt{\beta_1} = \frac{m_3}{\sigma^3}.$$

$\gamma_1 > 0$: positively skewed, $\gamma_1 = 0$: symmetric, $\gamma_1 < 0$: negatively skewed.

Solved Example

A small dataset of the time (in minutes) needed to repair five machines:

$$30,\ 35,\ 40,\ 45,\ 80.$$

Compute the mean, median, mode, and standard deviation. Then determine Pearson's skewness (median-based) and the moment coefficient of skewness γ_1.

Solution:

$$\bar{x} = \frac{30+35+40+45+80}{5} = \frac{230}{5} = 46 \text{ min}.$$

Ordered data: 30, 35, 40, 45, 80. Median = 40 min. No repeated values, so no mode (or all values are modes – use median-based Pearson).

$$\text{Variance } \sigma^2 = \frac{1}{5}\sum(x_i - 46)^2.$$

Deviations: $-16, -11, -6, -1, 34$; squares: $256, 121, 36, 1, 1156$; sum = 1570. $\sigma^2 = 1570/5 = 314$, $\sigma \approx 17.72$ min.

Pearson (median-based): $Sk_P = \frac{3(46-40)}{17.72} \approx \frac{18}{17.72} \approx 1.016$ (positive).

For moments, compute m_2 and m_3 about the mean. Deviations cubed: $(-16)^3 = -4096$, $(-11)^3 = -1331$, $(-6)^3 = -216$, $(-1)^3 = -1$, $34^3 = 39304$; sum = 33660.

$$m_3 = \frac{33660}{5} = 6732, \quad m_2 = \sigma^2 = 314.$$

$$\gamma_1 = \frac{m_3}{\sigma^3} = \frac{m_3}{m_2^{3/2}} = \frac{6732}{314^{1.5}}.$$

$314^{1.5} = 314 \times \sqrt{314} \approx 314 \times 17.72 \approx 5564$ (since $\sigma \approx 17.72$, $\sigma^3 \approx 17.72^3 \approx 5564$). Thus $\gamma_1 \approx 6732/5564 \approx 1.21$ (positive). Both methods confirm a strong right skew caused by the one long repair time of 80 min.

6 Kurtosis

Kurtosis describes the tailedness and peak sharpness of a distribution relative to a normal distribution. It is measured using the fourth central moment.

6.1 Types of Kurtosis

- **Mesokurtic**: $\beta_2 = 3$ (normal distribution).

- **Leptokurtic**: $\beta_2 > 3$, sharper peak, heavier tails (more frequent extreme values).
- **Platykurtic**: $\beta_2 < 3$, flatter peak, thinner tails.

6.2 Measure of Kurtosis

$$\beta_2 = \frac{m_4}{m_2^2}, \quad \text{where } m_4 = \frac{\sum f_i (x_i - \bar{x})^4}{N}.$$

Excess kurtosis is often reported:

$$\gamma_2 = \beta_2 - 3.$$

$\gamma_2 > 0$: leptokurtic, $\gamma_2 = 0$: mesokurtic, $\gamma_2 < 0$: platykurtic.

Solved Example

The second and fourth central moments of a dataset of 200 voltage measurements are $m_2 = 0.81$ and $m_4 = 2.43$. Determine the kurtosis and interpret.

Solution:

$$\beta_2 = \frac{m_4}{m_2^2} = \frac{2.43}{0.81^2} = \frac{2.43}{0.6561} \approx 3.704.$$

Excess kurtosis $\gamma_2 = 3.704 - 3 = 0.704 > 0$. The distribution is leptokurtic – sharper peak and heavier tails than a normal distribution, implying a higher probability of extreme voltage fluctuations. This could be critical in circuit protection design.

7 Summary Table of Descriptive Measures

Table 1: Summary of descriptive statistics

Aspect	**Measure**	**Key Formula**
Central Tendency	Arithmetic Mean	$\bar{x} = \frac{\sum f_i x_i}{N}$
	Median (grouped)	$L + \left(\frac{N/2 - F}{f_m}\right) w$
	Mode (grouped)	$L + \left(\frac{f_m - f_1}{2f_m - f_1 - f_2}\right) w$
Dispersion	Range	$X_{\max} - X_{\min}$
	Mean Deviation	$\frac{\sum f_i \lvert x_i - \bar{x} \rvert}{N}$
	Standard Deviation	$\sigma = \sqrt{\frac{\sum f_i (x_i - \bar{x})^2}{N}}$
	Coefficient of Variation	$\frac{\sigma}{\bar{x}} \times 100\%$
Skewness	Pearson's	$\frac{3(\text{Mean} - \text{Median})}{\sigma}$
	Moment γ_1	m_3/σ^3
Kurtosis	β_2	m_4/m_2^2
	Excess γ_2	$\beta_2 - 3$

Practice Problems

1. The following data are the compression strengths (in MPa) of 15 concrete cubes: 28.5, 31.2, 29.8, 33.1, 30.0, 32.4, 29.5, 31.0, 30.8, 32.0, 28.9, 31.5, 30.3, 33.0, 29.0. Compute mean, median, mode, range, mean deviation about the mean, standard deviation, and coefficient of variation.

2. The frequency distribution of the time to failure (in hours) of 80 electronic components is:

Time (hrs)	Components
0–50	5
50–100	12
100–150	28
150–200	22
200–250	10
250–300	3

   Using the step-deviation method, compute the mean, median, mode, standard deviation, and Pearson's coefficient of skewness.

3. For the repair time data in Example 5.2, compute the fourth central moment and the kurtosis β_2. (Hint: you already have the mean and deviations.)

4. A dataset of 500 network latency measurements yields $m_2 = 4.0$ ms^2 and $m_4 = 64.0$ ms^4. Find the kurtosis and discuss whether the latency distribution shows heavy tails.

5. Explain why standard deviation is preferred over mean deviation in quality control engineering.